

# IMO PHASE II LEVEL 2

## DOUBLE COUNTING

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**Double Counting:** The main idea is to count some quantity in two different ways. Often, the main difficulty of a double counting problem is figuring out the specific quantity to double count. It can be used to prove some equalities (or inequalities) or to find some bounds for the solution.

**Example 1.** Let  $p_n(k)$  be the number of permutations of the set  $S = \{1, 2, 3, \dots, n\}$  which have exactly  $k$  fixed points. Prove that

$$\sum_{k=0}^n kp_n(k) = n!.$$

(A permutation  $p$  is a bijective map from  $S \rightarrow S$ . An element  $s \in S$  is called a fixed point of  $p$  if  $p(s) = s$ .)

**Ans.** Let  $X$  be the collection of all pairs  $(s, p)$  where  $p \in \text{Sym}_n$  is a permutation with  $s \in S$  being one of its fixed points. We will count  $X$  in two ways:

LHS: For any  $p \in \text{Sym}_n$  with exactly  $k$  fixed points, it admits  $k$  such pairs in  $X$ . Hence  $|X| = \sum_{k=0}^n kp_n(k)$ .

RHS: For any  $s \in S$ , there are obviously  $(n-1)!$  permutations fix  $s$ . Hence  $|X| = n \cdot (n-1)!$ .

**Example 2.** (China TST 1992, problem 1) 16 students took part in a competition. All problems were multiple choice style. Each problem had four choices. It was said that any two students had at most one answer in common, find the maximum number of problems.

**Ans.** Let the students be  $S_1, S_2, \dots, S_{16}$  and the questions be  $Q_1, Q_2, \dots, Q_n$ . Define  $X$  as the number of triples  $(S_i, S_j, Q_k)$  where  $1 \leq i < j \leq 16$  such that  $S_i$  and  $S_j$  got the same answer for  $Q_k$ .

Because any two students have at most one answer in common, we have

$$X \leq 1 \cdot \binom{16}{2} = 120.$$

Now, we find a lower bound for  $X$  by analyzing each question individually.

For each  $Q_k$ , suppose  $a$  people answered "A" and so on. Since  $a + b + c + d = 16$ , Cauchy-Schwarz implies

$$\begin{aligned} \binom{a}{2} + \binom{b}{2} + \binom{c}{2} + \binom{d}{2} &= \frac{a^2 + b^2 + c^2 + d^2 - (a + b + c + d)}{2} \\ &\geq \frac{(a + b + c + d)^2}{8} - 8 = 24. \end{aligned}$$

Thus, there are at least 24 triples of the form  $(S_i, S_j, Q_k)$ . Now, symmetry implies

$$24n \leq X \leq 120 \implies n \leq 5.$$

**Problem 1.** A stage course is attended by  $n \geq 4$  students. The day before the final exam, each group of three students conspire against another student to throw him/her out of the exam. Prove that there is a student against whom there are at least  $\sqrt[3]{(n-1)(n-2)}$  conspirators.

**Problem 2.** (LMNas UGM 32 Final Round) Determine the smallest natural number  $n$  such that there exist sets  $A_1, A_2, \dots, A_n \subseteq \{1, 2, 3, \dots, n\}$  satisfying the following criteria:

1.  $i \notin A_i$  for every  $i = 1, 2, \dots, n$ ;
2.  $i \in A_j$  if and only if  $j \notin A_i$  for every distinct  $i$  and  $j$  where  $i, j \in \{1, 2, \dots, n\}$ ;
3.  $A_i \cap A_j$  is not an empty set for every  $i, j \in \{1, 2, \dots, n\}$ .

**Problem 3.** (Romanian Master Of Mathematics 2012) Given a positive integer  $n \geq 3$ , colour each cell of an  $n \times n$  square array with one of  $\lfloor (n+2)^2/3 \rfloor$  colours, each colour being used at least once. Prove that there is some  $1 \times 3$  or  $3 \times 1$  rectangular subarray whose three cells are coloured with three different colours.

**Problem 4.** Prove the identity:

$$\sum_{k=0}^n \binom{n}{k} \cdot \binom{n+k}{k} = \sum_{k=0}^n 2^k \cdot \binom{n}{k}^2$$

**Problem 5.** Prove that

$$\sum_{i=0}^n \sum_{j=0}^n \frac{(n+i+j)!}{n!i!j!} 3^{-i-j} = 3^n.$$

**Problem 6.**  $n > 2$  is an integer and  $p(> 2n-4)$  a prime. Let  $M$  be a set of  $n$  points on a plane. No three points are collinear. Integers are written on each point, satisfying following conditions.

1. There are exactly one integer which is  $0 \pmod{p}$ ;
2. If  $A, B, C$  are different three points of  $M$ , sum of integers on circumcircle of  $\triangle ABC$  is  $0 \pmod{p}$ .

Prove that all points of  $M$  are on a circle.

**Problem 7.** Let  $S$  be a set of  $n$  persons such that:

1. any person is acquainted to exactly  $k$  other persons in  $S$ ;
2. any two persons that are acquainted have exactly  $l$  common acquaintances in  $S$ ;
3. any two persons that are not acquainted have exactly  $m$  common acquaintances in  $S$ .

Prove that

$$m(n-k) - k(k-l) + k - m = 0.$$

**Problem 8.** (IMO 1989 Q3) Let  $n$  and  $k$  be positive integers and let  $S$  be a set of  $n$  points in the plane such that

1. no three points of  $S$  are collinear, and
2. for every point  $P$  of  $S$  there are at least  $k$  points of  $S$  equidistant from  $P$ .

Prove that

$$k < \frac{1}{2} + \sqrt{2 \cdot n}.$$

**Problem 9.** (IMO 1998 Q2) In a contest, there are  $m$  candidates and  $n$  judges, where  $n \geq 3$  is an odd integer. Each candidate is evaluated by each judge as either pass or fail. Suppose that each pair of judges agrees on at most  $k$  candidates. Prove that

$$\frac{k}{m} \geq \frac{n-1}{2n}.$$

**Problem 10.** (Iran Olympiad) A school has  $n$  students. Each student can participate in any number of classes that he wants. Every class has at least two students participating in it, and if two different classes have at least two common students, then the number of students in these two classes is different. Prove that the number of classes is not greater than  $(n-1)^2$ .

**Problem 11.** (Russia 1999) In a class, every boy knows at least one girl. Prove that there is a group containing at least half of the students such that each boy in the group knows an odd number of girls in the group.